

## Qualification Pack



# Junior Engineer - Automatic Fare Collection and Telecom Systems (Metro Rail)

QP Code: DMR/LSC/Q0202

Version: 1.0

NSQF Level: 4.5

Delhi Metro Rail Corporation Limited || M6FR+8V5, Depot Entry Road, Metro Vihar, Shastri Park, Shahdara, Delhi

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## Qualification Pack

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## Qualification Pack

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## Qualification Pack

# DMR/LSC/Q0202: Junior Engineer - Automatic Fare Collection and Telecom Systems (Metro Rail)

## Brief Job Description

A Junior Engineer - Automatic Fare Collection and Telecom Systems (Metro Rail) is responsible for performing basic signalling operations, installing, operating, and maintaining Telecom, and Automatic Fare Collection (AFC) Systems. The individual ensures system efficiency, compliance with safety standards, and effective team coordination to maintain uninterrupted metro operations.

## Personal Attributes

The role requires strong analytical skills, attention to detail, effective communication, and organizational skills.

## Applicable National Occupational Standards (NOS)

### Compulsory NOS:

1. [DMR/LSC/N9901: Implement metro operations training and safety procedures](#)
2. [DMR/LSC/N0203: Conduct basic signalling operations](#)
3. [DMR/LSC/N0204: Execute telecom systems installation, operation, and maintenance](#)
4. [DMR/LSC/N0205: Operate and Monitor Automatic Fare Collection \(AFC\) Systems](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

## Qualification Pack (QP) Parameters

|                                      |                        |
|--------------------------------------|------------------------|
| <b>Sector</b>                        | Logistics              |
| <b>Sub-Sector</b>                    | Transport (Metro Rail) |
| <b>Occupation</b>                    | Signalling & Telecom   |
| <b>Country</b>                       | India                  |
| <b>NSQF Level</b>                    | 4.5                    |
| <b>Credits</b>                       | 18                     |
| <b>Aligned to NCO/ISCO/ISIC Code</b> | NCO-2015/2153.9900     |

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|                                                           |                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Minimum Educational Qualification &amp; Experience</b> | Diploma ( 3 years Engineering Diploma* / equivalent in Electronics, Computer Engg., Information Technology, Electronics & Communications, Electronics & Communications Industry Integrated, Electrical & Electronics, Electronics / Microprocessor, Electronics & Telecommunications, Instrumentation Technology, Electronics and Instrumentation Engg. or in relevant field) |
| <b>Minimum Level of Education for Training in School</b>  | Not Applicable                                                                                                                                                                                                                                                                                                                                                                |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Minimum Job Entry Age</b>                              | 18 Years                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Next Review Date</b>                                   | 08/05/2028                                                                                                                                                                                                                                                                                                                                                                    |
| <b>NSQC Approval Date</b>                                 | 08/05/2025                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Version</b>                                            | 1.0                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Reference code on NQR</b>                              | QG-4.5-TW-04143-2025-V1-DMRCA                                                                                                                                                                                                                                                                                                                                                 |
| <b>NQR Version</b>                                        | 1.0                                                                                                                                                                                                                                                                                                                                                                           |

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### DMR/LSC/N9901: Implement metro operations training and safety procedures

#### Description

This unit focuses on implementing structured training and safety measures for metro operations, ensuring technical knowledge, system readiness, and emergency preparedness.

#### Scope

The scope covers the following :

- Training Module Overview
- Safety Overview
- Store Management and Procurement
- Overview of Telecom, Signaling, and Track Systems
- Overview of Traction, SCADA, and E&M Systems
- OCC and Rolling Stock Overview
- Operations and Customer Care
- Practical Visits
- First Aid Training
- Fire Fighting Systems
- Security in Metro Systems
- Metro Rail General Rules 2020
- Safety Circulars and Contingency Plans

#### Elements and Performance Criteria

##### *Training Module Overview*

To be competent, the user/individual on the job must be able to:

- PC1.** follow schedules for weekly activities and guidelines on punctuality, discipline, and conduct.
- PC2.** conduct identity verification and maintain attendance records
- PC3.** ensure adherence to training timelines and program deliverables

##### *Safety Overview*

To be competent, the user/individual on the job must be able to:

- PC4.** identify the role of the safety department and follow measures for personal, passenger, and system safety
- PC5.** use appropriate Personal Protective Equipment (PPE) in all designated areas
- PC6.** follow the hazard identification and risk assessment processes

##### *Store Management and Procurement*

To be competent, the user/individual on the job must be able to:

- PC7.** perform store management activities, including stock handling, documentation, and interdepartmental coordination for procurement.
- PC8.** monitor inventory levels and ensure timely requisitions for required materials
- PC9.** maintain records of procurement processes and vendor communication

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### *Overview of Telecom, Signaling, and Track Systems*

To be competent, the user/individual on the job must be able to:

- PC10.** identify telecom components (AFC, CCTV, PIDS, radio systems, clocks) and perform basic functional checks
- PC11.** identify signaling systems (CATC, CBTC) and observe/report signal functioning during visits.
- PC12.** identify P-Way components (rails, sleepers, crossings) and report track conditions through basic patrolling tasks
- PC13.** report malfunctions in telecom, signaling, or track systems using the correct reporting protocols

### *Overview of Traction, SCADA, and E&M Systems*

To be competent, the user/individual on the job must be able to:

- PC14.** identify traction power and SCADA components and interpret sectioning diagrams
- PC15.** identify E&M systems (RSS, ASS, DG sets, UPS) and perform basic checks to understand their functionality
- PC16.** report unusual system performance and assist in routine inspections

### *OCC and Rolling Stock Overview*

To be competent, the user/individual on the job must be able to:

- PC17.** familiarize with OCC systems, tools, and the role of OCC controllers
- PC18.** identify rolling stock components and their maintenance procedures
- PC19.** report rolling stock irregularities observed during operations

### *Operations and Customer Care*

To be competent, the user/individual on the job must be able to:

- PC20.** familiarize with station operations, revenue management, and crew control functions
- PC21.** identify public complaint channels, possible causes, and resolve complaints using proper processes
- PC22.** assist in managing passenger inquiries and ensuring customer satisfaction

### *Practical Visits*

To be competent, the user/individual on the job must be able to:

- PC23.** conduct visits to OCC, stations, depots, and tunnels to observe layouts, facilities, and operations
- PC24.** document key observations during practical visits for knowledge reinforcement
- PC25.** identify operational challenges during visits and suggest improvement areas

### *First Aid Training*

To be competent, the user/individual on the job must be able to:

- PC26.** identify the role and scope of first aid, including key human body functions (e.g., circulatory, respiratory systems)
- PC27.** follow appropriate techniques for treating bleeding, fractures, burns, shock, and unconsciousness
- PC28.** perform artificial respiration, CPR, and safe transportation of injured persons using stretchers
- PC29.** participate in simulated first aid scenarios to enhance preparedness

### *Fire Fighting Systems*

To be competent, the user/individual on the job must be able to:



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- PC30.** identify elements of combustion, classes of fire, and appropriate extinguishing methods
- PC31.** operate active/passive fire protection systems, including alarms, fire extinguishers, and fire hydrants
- PC32.** conduct practical drills and identify fire suppression equipment at metro stations
- PC33.** respond to fire incidents promptly, ensuring passenger and system safety

### *Security in Metro Systems*

To be competent, the user/individual on the job must be able to:

- PC34.** identify security threats, red alert situations, and follow protocols for disaster management
- PC35.** handle theft cases, manage abandoned persons, and lodge FIRs as per protocol
- PC36.** coordinate with security personnel to mitigate security breaches effectively

### *Metro Rail General Rules 2020*

To be competent, the user/individual on the job must be able to:

- PC37.** interpret key provisions under Metro Rail General Rules 2020, including signals, train speed limits, and safety protocols
- PC38.** follow operational procedures for maintenance, station control duties, and driverless train systems
- PC39.** ensure compliance with rules for safe and efficient train operations

### *Safety Circulars and Contingency Plans*

To be competent, the user/individual on the job must be able to:

- PC40.** implement procedures for the imposition of TSR, axle counter reset, Train ID management, and Computer Aided Train Supervision (CATS) operation
- PC41.** follow guidelines for engineering possession and maintenance activities during non-revenue hours, including hot work, solar system testing/repair, and role-play exercises
- PC42.** ensure compliance with procedures for stopping trains at platforms and during operational disruptions, such as earthquakes, high winds, and floods
- PC43.** manage degraded mode operations, including train movement during interlocking failures
- PC44.** oversee the issue, revision, updating, and deletion of SOPs, SWO/DWO, and timetable trials for new sections
- PC45.** facilitate the initiation of revenue train services, including pilot train movement, ATO/UTO/ATB mode operations, and AFTC functionality checks
- PC46.** execute procedures for negotiating neutral sections, managing pantograph failures, and monitoring Panto-OHE interface cameras
- PC47.** oversee train stabling, depot entry/exit, train movement, and fitness checks
- PC48.** conduct emergency evacuation using UG facilities, mid-tunnel shafts, and walkways for detainment of passengers due to technical failure, explosion, or fire
- PC49.** operate and manage non-ATP vehicles such as RGM and CMV/CTMC and address incidents like rail fractures, weld failures, rollback scenarios, radio communication issues and resetting of passenger alarms in train
- PC50.** coordinate vehicle movement on link lines and rescue immobilized trains
- PC51.** respond to fire/smoke incidents, including UG train fires, ensuring passenger and staff safety
- PC52.** operate ETS/BLS plungers, remove foreign objects like kites from OHE, and follow OHE safety guidelines



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### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of punctuality, discipline, and structured training
- KU2.** the procedures for trainee verification and attendance record maintenance
- KU3.** the role of time management in achieving training milestones
- KU4.** the role of the safety department and station/depot safety protocols
- KU5.** the use and importance of PPE
- KU6.** the basics of hazard identification, risk management, and accident reporting
- KU7.** the basics of stores management (inventory control, documentation)
- KU8.** the procurement processes and coordination across departments
- KU9.** the procedures for vendor management and maintaining procurement transparency
- KU10.** the telecom components, including AFC, CCTV, clocks, and communication systems
- KU11.** the basics of signaling systems (CATC, CBTC) and reporting unusual signal observations
- KU12.** the key P-Way components, track patrolling, and reporting track conditions
- KU13.** the common malfunctions in telecom, signaling, and track systems
- KU14.** the traction power systems and sectioning diagrams
- KU15.** the importance of SCADA and its functionalities
- KU16.** the role of E&M systems (DG sets, UPS) and their use during emergencies
- KU17.** the key preventive maintenance checks for E&M systems
- KU18.** the OCC systems, tools, and controller responsibilities
- KU19.** the rolling stock components, features, and basic maintenance
- KU20.** the reporting and addressing of common rolling stock issues
- KU21.** the station operations, revenue management, and crew coordination
- KU22.** the causes of public complaints and grievance resolution processes
- KU23.** the techniques for effective customer service and passenger engagement
- KU24.** the observations and learning outcomes from OCC, station, depot, and tunnel visits
- KU25.** the identifying real-world challenges during visits and linking theory to practice
- KU26.** the basic principles of first aid and key human body functions
- KU27.** the management of bleeding, fractures, burns, and unconsciousness
- KU28.** the artificial respiration, CPR, and safe patient transportation
- KU29.** the importance of first aid preparedness during emergencies
- KU30.** the elements of combustion, fire classifications, and extinguishing methods
- KU31.** the principles of passive/active fire protection systems
- KU32.** the practical usage of fire extinguishers, hydrants, and drills
- KU33.** the emergency response procedures during fire incidents
- KU34.** the security protocols, red alert situations, and theft management processes
- KU35.** the protocols for managing abandoned persons and disaster situations
- KU36.** the role of security personnel in ensuring passenger and asset safety
- KU37.** the key terminologies, signal usage, and operational protocols

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- KU38.** the rules for maintenance works, train operations, and driverless train systems
- KU39.** the role of regulatory compliance in metro rail safety
- KU40.** the safety circulars and contingency plans for fire/security incidents
- KU41.** the importance of proactive contingency planning for emergency scenarios.
- KU42.** the importance role-playing and drills to improve safety and response performance
- KU43.** the procedures for TSR imposition, axle counter reset, Train ID management, and CATS operations
- KU44.** the engineering possession protocols for maintenance tasks during non-revenue hours, including solar system testing and hot work
- KU45.** the guidelines for safely halting train operations during emergencies, natural disasters, or technical failures
- KU46.** the steps for managing degraded mode operations and interlocking failures
- KU47.** the SOP and SWO/DWO lifecycle management, including revisions and timetable trials for new sections
- KU48.** the ATO, UTO, and ATB mode operations, emphasizing operational readiness and AFTC checks
- KU49.** the procedures for neutral section negotiation, pantograph damage management, and monitoring Panto-OHE interfaces using cameras
- KU50.** the best practices for train stabling, depot operations, and train fitness verification
- KU51.** the emergency evacuation protocols, focusing on passenger safety in UG and mid-tunnel situations
- KU52.** the guidelines for operating non-ATP vehicles, handling rail fractures, weld failures, and resetting alarms during rollbacks
- KU53.** the vehicle operations on link lines and immobilized train rescue procedures
- KU54.** the protocols for fire/smoke incidents, particularly UG fires, and ensuring safety during such incidents
- KU55.** the operation of ETS/BLS plungers and removal of foreign objects from OHE systems, along with OHE safety practices

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** convey clear and concise information to teams, junior engineers, and customers
- GS2.** effectively prioritize tasks and adhere to training schedules and operational timelines
- GS3.** identify operational challenges and propose practical solutions during training or real-time operations
- GS4.** assess system performance, identify malfunctions, and report anomalies accurately
- GS5.** work cohesively with peers, instructors, and junior engineers to meet training objectives
- GS6.** address passenger queries and manage grievances professionally
- GS7.** proactively identify risks, follow safety protocols, and contribute to hazard-free operations
- GS8.** adapt to diverse training environments, system layouts, and evolving operational scenarios
- GS9.** accurately maintain logs, records, and reports for training and practical activities

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- GS10.** monitor systems, equipment, and processes, identifying areas requiring intervention or improvement

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### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Training Module Overview</i>                                                                                                               | 2            | 2               | -             | 1          |
| <b>PC1.</b> follow schedules for weekly activities and guidelines on punctuality, discipline, and conduct.                                    | -            | -               | -             | -          |
| <b>PC2.</b> conduct identity verification and maintain attendance records                                                                     | -            | -               | -             | -          |
| <b>PC3.</b> ensure adherence to training timelines and program deliverables                                                                   | -            | -               | -             | -          |
| <i>Safety Overview</i>                                                                                                                        | 2            | 2               | -             | 1          |
| <b>PC4.</b> identify the role of the safety department and follow measures for personal, passenger, and system safety                         | -            | -               | -             | -          |
| <b>PC5.</b> use appropriate Personal Protective Equipment (PPE) in all designated areas                                                       | -            | -               | -             | -          |
| <b>PC6.</b> follow the hazard identification and risk assessment processes                                                                    | -            | -               | -             | -          |
| <i>Store Management and Procurement</i>                                                                                                       | 2            | 2               | -             | 1          |
| <b>PC7.</b> perform store management activities, including stock handling, documentation, and interdepartmental coordination for procurement. | -            | -               | -             | -          |
| <b>PC8.</b> monitor inventory levels and ensure timely requisitions for required materials                                                    | -            | -               | -             | -          |
| <b>PC9.</b> maintain records of procurement processes and vendor communication                                                                | -            | -               | -             | -          |
| <i>Overview of Telecom, Signaling, and Track Systems</i>                                                                                      | 2            | 2               | -             | 1          |
| <b>PC10.</b> identify telecom components (AFC, CCTV, PIDS, radio systems, clocks) and perform basic functional checks                         | -            | -               | -             | -          |
| <b>PC11.</b> identify signaling systems (CATC, CBTC) and observe/report signal functioning during visits.                                     | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> identify P-Way components (rails, sleepers, crossings) and report track conditions through basic patrolling tasks | -            | -               | -             | -          |
| <b>PC13.</b> report malfunctions in telecom, signaling, or track systems using the correct reporting protocols                 | -            | -               | -             | -          |
| <i>Overview of Traction, SCADA, and E&amp;M Systems</i>                                                                        | <b>2</b>     | <b>2</b>        | -             | <b>1</b>   |
| <b>PC14.</b> identify traction power and SCADA components and interpret sectioning diagrams                                    | -            | -               | -             | -          |
| <b>PC15.</b> identify E&M systems (RSS, ASS, DG sets, UPS) and perform basic checks to understand their functionality          | -            | -               | -             | -          |
| <b>PC16.</b> report unusual system performance and assist in routine inspections                                               | -            | -               | -             | -          |
| <i>OCC and Rolling Stock Overview</i>                                                                                          | <b>2</b>     | <b>2</b>        | -             | <b>1</b>   |
| <b>PC17.</b> familiarize with OCC systems, tools, and the role of OCC controllers                                              | -            | -               | -             | -          |
| <b>PC18.</b> identify rolling stock components and their maintenance procedures                                                | -            | -               | -             | -          |
| <b>PC19.</b> report rolling stock irregularities observed during operations                                                    | -            | -               | -             | -          |
| <i>Operations and Customer Care</i>                                                                                            | <b>2</b>     | <b>2</b>        | -             | <b>1</b>   |
| <b>PC20.</b> familiarize with station operations, revenue management, and crew control functions                               | -            | -               | -             | -          |
| <b>PC21.</b> identify public complaint channels, possible causes, and resolve complaints using proper processes                | -            | -               | -             | -          |
| <b>PC22.</b> assist in managing passenger inquiries and ensuring customer satisfaction                                         | -            | -               | -             | -          |
| <i>Practical Visits</i>                                                                                                        | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC23.</b> conduct visits to OCC, stations, depots, and tunnels to observe layouts, facilities, and operations               | -            | -               | -             | -          |

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| Assessment Criteria for Outcomes                                                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC24.</b> document key observations during practical visits for knowledge reinforcement                                         | -            | -               | -             | -          |
| <b>PC25.</b> identify operational challenges during visits and suggest improvement areas                                           | -            | -               | -             | -          |
| <i>First Aid Training</i>                                                                                                          | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC26.</b> identify the role and scope of first aid, including key human body functions (e.g., circulatory, respiratory systems) | -            | -               | -             | -          |
| <b>PC27.</b> follow appropriate techniques for treating bleeding, fractures, burns, shock, and unconsciousness                     | -            | -               | -             | -          |
| <b>PC28.</b> perform artificial respiration, CPR, and safe transportation of injured persons using stretchers                      | -            | -               | -             | -          |
| <b>PC29.</b> participate in simulated first aid scenarios to enhance preparedness                                                  | -            | -               | -             | -          |
| <i>Fire Fighting Systems</i>                                                                                                       | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC30.</b> identify elements of combustion, classes of fire, and appropriate extinguishing methods                               | -            | -               | -             | -          |
| <b>PC31.</b> operate active/passive fire protection systems, including alarms, fire extinguishers, and fire hydrants               | -            | -               | -             | -          |
| <b>PC32.</b> conduct practical drills and identify fire suppression equipment at metro stations                                    | -            | -               | -             | -          |
| <b>PC33.</b> respond to fire incidents promptly, ensuring passenger and system safety                                              | -            | -               | -             | -          |
| <i>Security in Metro Systems</i>                                                                                                   | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC34.</b> identify security threats, red alert situations, and follow protocols for disaster management                         | -            | -               | -             | -          |
| <b>PC35.</b> handle theft cases, manage abandoned persons, and lodge FIRs as per protocol                                          | -            | -               | -             | -          |
| <b>PC36.</b> coordinate with security personnel to mitigate security breaches effectively                                          | -            | -               | -             | -          |

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| Assessment Criteria for Outcomes                                                                                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Metro Rail General Rules 2020</i>                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC37.</b> interpret key provisions under Metro Rail General Rules 2020, including signals, train speed limits, and safety protocols                                                  | -            | -               | -             | -          |
| <b>PC38.</b> follow operational procedures for maintenance, station control duties, and driverless train systems                                                                        | -            | -               | -             | -          |
| <b>PC39.</b> ensure compliance with rules for safe and efficient train operations                                                                                                       | -            | -               | -             | -          |
| <i>Safety Circulars and Contingency Plans</i>                                                                                                                                           | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC40.</b> implement procedures for the imposition of TSR, axle counter reset, Train ID management, and Computer Aided Train Supervision (CATS) operation                             | -            | -               | -             | -          |
| <b>PC41.</b> follow guidelines for engineering possession and maintenance activities during non-revenue hours, including hot work, solar system testing/repair, and role-play exercises | -            | -               | -             | -          |
| <b>PC42.</b> ensure compliance with procedures for stopping trains at platforms and during operational disruptions, such as earthquakes, high winds, and floods                         | -            | -               | -             | -          |
| <b>PC43.</b> manage degraded mode operations, including train movement during interlocking failures                                                                                     | -            | -               | -             | -          |
| <b>PC44.</b> oversee the issue, revision, updating, and deletion of SOPs, SWO/DWO, and timetable trials for new sections                                                                | -            | -               | -             | -          |
| <b>PC45.</b> facilitate the initiation of revenue train services, including pilot train movement, ATO/UTO/ATB mode operations, and AFTC functionality checks                            | -            | -               | -             | -          |
| <b>PC46.</b> execute procedures for negotiating neutral sections, managing pantograph failures, and monitoring Panto-OHE interface cameras                                              | -            | -               | -             | -          |



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| Assessment Criteria for Outcomes                                                                                                                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC47.</b> oversee train stabling, depot entry/exit, train movement, and fitness checks                                                                                                                                     | -            | -               | -             | -          |
| <b>PC48.</b> conduct emergency evacuation using UG facilities, mid-tunnel shafts, and walkways for detrainment of passengers due to technical failure, explosion, or fire                                                     | -            | -               | -             | -          |
| <b>PC49.</b> operate and manage non-ATP vehicles such as RGM and CMV/CTMC and address incidents like rail fractures, weld failures, rollback scenarios, radio communication issues and resetting of passenger alarms in train | -            | -               | -             | -          |
| <b>PC50.</b> coordinate vehicle movement on link lines and rescue immobilized trains                                                                                                                                          | -            | -               | -             | -          |
| <b>PC51.</b> respond to fire/smoke incidents, including UG train fires, ensuring passenger and staff safety                                                                                                                   | -            | -               | -             | -          |
| <b>PC52.</b> operate ETS/BLS plungers, remove foreign objects like kites from OHE, and follow OHE safety guidelines                                                                                                           | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                              | <b>20</b>    | <b>20</b>       | <b>-</b>      | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                           |
|----------------------------|-----------------------------------------------------------|
| <b>NOS Code</b>            | DMR/LSC/N9901                                             |
| <b>NOS Name</b>            | Implement metro operations training and safety procedures |
| <b>Sector</b>              | Logistics                                                 |
| <b>Sub-Sector</b>          |                                                           |
| <b>Occupation</b>          | Foundation – Metro (Rail)                                 |
| <b>NSQF Level</b>          | 4.5                                                       |
| <b>Credits</b>             | 3                                                         |
| <b>Version</b>             | 1.0                                                       |
| <b>Last Reviewed Date</b>  | 18/02/2025                                                |
| <b>Next Review Date</b>    | 18/02/2028                                                |
| <b>NSQC Clearance Date</b> | 18/02/2025                                                |

## Qualification Pack

### DMR/LSC/N0203: Conduct basic signalling operations

#### Description

This unit covers the practical skills and competencies required to execute, monitor, and maintain signalling systems and their interfacing components in metro rail operations.

#### Scope

The scope covers the following :

- General/PSD
- Interlocking
- Point Machine
- Signal
- Track Detection Devices
- Automatic Train Supervision (ATS)
- Automatic Train Protection (ATP)
- Relays
- Duties of S&TC Junior Engineers
- Telecom
- Civil
- Traction
- E&M
- Rolling Stock
- Visits

#### Elements and Performance Criteria

##### General/PSD

To be competent, the user/individual on the job must be able to:

- PC1.** familiarize with the role of the S&T department in metro operations
- PC2.** monitor signalling systems to ensure smooth metro operations
- PC3.** identify and use signalling terminologies during maintenance and operation tasks
- PC4.** inspect various S&T systems used in DMRC for proper functioning
- PC5.** operate the Communication-Based Train Control (CBTC) system following the applicable procedure
- PC6.** test the integration and functionality of Platform Screen Doors (PSD) in metro stations
- PC7.** apply the system of working protocols in signalling operations
- PC8.** check signalling operations within the depot environment
- PC9.** install and interpret markers and indicators for train operations
- PC10.** analyze overlap, isolation, and sighting distances as per operational requirements
- PC11.** check for Schedule of Dimensions (SOD) compliance for signalling-related installations
- PC12.** prepare and interpret control tables for signalling operations

##### Interlocking

## Qualification Pack

To be competent, the user/individual on the job must be able to:

**PC13.** execute interlocking procedures to ensure safe train operations

**PC14.** compare mechanical, RRI, SSI, and CBI interlocking systems for specific operational needs

### *Point Machine*

To be competent, the user/individual on the job must be able to:

**PC15.** operate and calibrate point machines for optimal performance

**PC16.** examine points and crossings to ensure alignment and safety during train movement

### *Signal*

To be competent, the user/individual on the job must be able to:

**PC17.** install, inspect, and maintain signals from semaphore to colour light systems

### *Track Detection Devices*

To be competent, the user/individual on the job must be able to:

**PC18.** install and maintain train detection devices, including track circuits and axle counters

**PC19.** operate track circuits and axle counters for detection

### *Automatic Train Supervision (ATS)*

To be competent, the user/individual on the job must be able to:

**PC20.** operate Automatic Train Supervision (ATS) systems for CBTC and Non-CBTC environments

**PC21.** simulate train operations on CBTC and Non-CBTC ATS systems to understand functionalities

### *Automatic Train Protection (ATP)*

To be competent, the user/individual on the job must be able to:

**PC22.** perform operational checks for different grades of ATP to ensure safe train movement

### *Relays*

To be competent, the user/individual on the job must be able to:

**PC23.** install, test, and maintain relays used in signalling systems

### *Duties of S&TC Junior Engineers*

To be competent, the user/individual on the job must be able to:

**PC24.** carry out general maintenance activities while adhering to safety protocols

**PC25.** coordinate with the S&TC staff to ensure compliance with operational standards

### *Telecom*

To be competent, the user/individual on the job must be able to:

**PC26.** test telecom systems for proper interfacing with signalling equipment

**PC27.** operate emergency communication systems in coordination with signalling functions

**PC28.** inspect fiber optic transmission systems for faults and ensure smooth communication

**PC29.** set up and monitor networking systems relevant to signalling operations

### *Civil*

To be competent, the user/individual on the job must be able to:

**PC30.** coordinate with P-way teams to ensure proper interfacing with signalling systems

### *Traction*

To be competent, the user/individual on the job must be able to:

**PC31.** inspect and test OHE systems interfacing with signalling for operational efficiency

**PC32.** apply safety practices for OHE operations to ensure signalling reliability

## Qualification Pack

### *E&M*

To be competent, the user/individual on the job must be able to:

**PC33.** monitor the interfacing of lifts and escalators with signalling systems

**PC34.** inspect and test SCADA and TVS systems for proper interfacing with signalling equipment

### *Rolling Stock*

To be competent, the user/individual on the job must be able to:

**PC35.** verify the integration of rolling stock with CBTC and Non-CBTC systems for compatibility

**PC36.** operate train CCTV systems to ensure effective interfacing with signalling systems

### *Visits*

To be competent, the user/individual on the job must be able to:

**PC37.** check equipment in the signalling equipment room to ensure compliance with the applicable standards

**PC38.** observe and report operations at the depot control center for process optimization

**PC39.** monitor signalling functions at stations to ensure reliability and safety

**PC40.** analyze the operational processes during visits to FMC and Central ATS to identify improvement opportunities

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** the basics of metro signalling concepts, CBTC, and S&T systems

**KU2.** the principles and essentials of various interlocking systems (mechanical, RRI, SSI, CBI)

**KU3.** the types, working mechanisms, and integration of point machines with signalling systems

**KU4.** the principles of operation for track detection devices like track circuits and axle counters

**KU5.** the evolution of signal systems and their practical applications

**KU6.** the features, advantages, and operational aspects of ATS and ATP

**KU7.** the working, preventive maintenance, and troubleshooting of relays

**KU8.** the basics of fiber optics, emergency communication, and networking

**KU9.** the system interfacing of P-way, traction, E&M, and rolling stock with signalling

**KU10.** the benefits of field observations from visits to control centers, stations, and depots

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** identify and promptly resolve faults in signalling and interfacing systems

**GS2.** collaborate with cross-functional teams for smooth operations

**GS3.** oversee day-to-day signalling operations for compliance and efficiency

**GS4.** document maintenance activities, faults, and solutions systematically

**GS5.** ensure compliance with safety regulations in all tasks

**GS6.** prioritize and schedule maintenance and operational activities effectively

Logo  
Not  
Available



### Qualification Pack

- GS7.** assess system performance and suggest improvements
- GS8.** relay technical information clearly to team members and stakeholders
- GS9.** follow updates on advancements in metro signalling technologies

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>General/PSD</i>                                                                                         | <b>6</b>     | <b>6</b>        | -             | <b>3</b>   |
| <b>PC1.</b> familiarize with the role of the S&T department in metro operations                            | -            | -               | -             | -          |
| <b>PC2.</b> monitor signalling systems to ensure smooth metro operations                                   | -            | -               | -             | -          |
| <b>PC3.</b> identify and use signalling terminologies during maintenance and operation tasks               | -            | -               | -             | -          |
| <b>PC4.</b> inspect various S&T systems used in DMRC for proper functioning                                | -            | -               | -             | -          |
| <b>PC5.</b> operate the Communication-Based Train Control (CBTC) system following the applicable procedure | -            | -               | -             | -          |
| <b>PC6.</b> test the integration and functionality of Platform Screen Doors (PSD) in metro stations        | -            | -               | -             | -          |
| <b>PC7.</b> apply the system of working protocols in signalling operations                                 | -            | -               | -             | -          |
| <b>PC8.</b> check signalling operations within the depot environment                                       | -            | -               | -             | -          |
| <b>PC9.</b> install and interpret markers and indicators for train operations                              | -            | -               | -             | -          |
| <b>PC10.</b> analyze overlap, isolation, and sighting distances as per operational requirements            | -            | -               | -             | -          |
| <b>PC11.</b> check for Schedule of Dimensions (SOD) compliance for signalling-related installations        | -            | -               | -             | -          |
| <b>PC12.</b> prepare and interpret control tables for signalling operations                                | -            | -               | -             | -          |
| <i>Interlocking</i>                                                                                        | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC13.</b> execute interlocking procedures to ensure safe train operations                               | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC14.</b> compare mechanical, RRI, SSI, and CBI interlocking systems for specific operational needs | -            | -               | -             | -          |
| <i>Point Machine</i>                                                                                   | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC15.</b> operate and calibrate point machines for optimal performance                              | -            | -               | -             | -          |
| <b>PC16.</b> examine points and crossings to ensure alignment and safety during train movement         | -            | -               | -             | -          |
| <i>Signal</i>                                                                                          | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC17.</b> install, inspect, and maintain signals from semaphore to colour light systems             | -            | -               | -             | -          |
| <i>Track Detection Devices</i>                                                                         | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC18.</b> install and maintain train detection devices, including track circuits and axle counters  | -            | -               | -             | -          |
| <b>PC19.</b> operate track circuits and axle counters for detection                                    | -            | -               | -             | -          |
| <i>Automatic Train Supervision (ATS)</i>                                                               | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC20.</b> operate Automatic Train Supervision (ATS) systems for CBTC and Non-CBTC environments      | -            | -               | -             | -          |
| <b>PC21.</b> simulate train operations on CBTC and Non-CBTC ATS systems to understand functionalities  | -            | -               | -             | -          |
| <i>Automatic Train Protection (ATP)</i>                                                                | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC22.</b> perform operational checks for different grades of ATP to ensure safe train movement      | -            | -               | -             | -          |
| <i>Relays</i>                                                                                          | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC23.</b> install, test, and maintain relays used in signalling systems                             | -            | -               | -             | -          |
| <i>Duties of S&amp;TC Junior Engineers</i>                                                             | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC24.</b> carry out general maintenance activities while adhering to safety protocols               | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC25.</b> coordinate with the S&TC staff to ensure compliance with operational standards           | -            | -               | -             | -          |
| <i>Telecom</i>                                                                                        | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC26.</b> test telecom systems for proper interfacing with signalling equipment                    | -            | -               | -             | -          |
| <b>PC27.</b> operate emergency communication systems in coordination with signalling functions        | -            | -               | -             | -          |
| <b>PC28.</b> inspect fiber optic transmission systems for faults and ensure smooth communication      | -            | -               | -             | -          |
| <b>PC29.</b> set up and monitor networking systems relevant to signalling operations                  | -            | -               | -             | -          |
| <i>Civil</i>                                                                                          | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC30.</b> coordinate with P-way teams to ensure proper interfacing with signalling systems         | -            | -               | -             | -          |
| <i>Traction</i>                                                                                       | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC31.</b> inspect and test OHE systems interfacing with signalling for operational efficiency      | -            | -               | -             | -          |
| <b>PC32.</b> apply safety practices for OHE operations to ensure signalling reliability               | -            | -               | -             | -          |
| <i>E&amp;M</i>                                                                                        | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC33.</b> monitor the interfacing of lifts and escalators with signalling systems                  | -            | -               | -             | -          |
| <b>PC34.</b> inspect and test SCADA and TVS systems for proper interfacing with signalling equipment  | -            | -               | -             | -          |
| <i>Rolling Stock</i>                                                                                  | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC35.</b> verify the integration of rolling stock with CBTC and Non-CBTC systems for compatibility | -            | -               | -             | -          |
| <b>PC36.</b> operate train CCTV systems to ensure effective interfacing with signalling systems       | -            | -               | -             | -          |
| <i>Visits</i>                                                                                         | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC37.</b> check equipment in the signalling equipment room to ensure compliance with the applicable standards          | -            | -               | -             | -          |
| <b>PC38.</b> observe and report operations at the depot control center for process optimization                           | -            | -               | -             | -          |
| <b>PC39.</b> monitor signalling functions at stations to ensure reliability and safety                                    | -            | -               | -             | -          |
| <b>PC40.</b> analyze the operational processes during visits to FMC and Central ATS to identify improvement opportunities | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                          | <b>20</b>    | <b>20</b>       | <b>-</b>      | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                     |
|----------------------------|-------------------------------------|
| <b>NOS Code</b>            | DMR/LSC/N0203                       |
| <b>NOS Name</b>            | Conduct basic signalling operations |
| <b>Sector</b>              | Logistics                           |
| <b>Sub-Sector</b>          |                                     |
| <b>Occupation</b>          | Signalling & Telecom                |
| <b>NSQF Level</b>          | 4.5                                 |
| <b>Credits</b>             | 3                                   |
| <b>Version</b>             | 1.0                                 |
| <b>Last Reviewed Date</b>  | 08/05/2025                          |
| <b>Next Review Date</b>    | 08/05/2028                          |
| <b>NSQC Clearance Date</b> | 08/05/2025                          |

## Qualification Pack

# DMR/LSC/N0204: Execute telecom systems installation, operation, and maintenance

## Description

This unit covers the execution, monitoring, and maintenance of telecom systems for efficient operation, system integrity, and optimal performance.

## Scope

The scope covers the following :

- Introduction to Telecom Systems
- Communication Media
- Fiber Optical Transmission System
- Radio System
- Telephone System
- PIDS & PAS
- Closed Circuit Television System (CCTV)
- Clock System
- Networking System

## Elements and Performance Criteria

### *Introduction to Telecom Systems*

To be competent, the user/individual on the job must be able to:

- PC1.** interpret the telecom system architecture and communication networks
- PC2.** familiarize with the telecom system and components through a station visit

### *Communication Media*

To be competent, the user/individual on the job must be able to:

- PC3.** identify different types of electrical cables (e.g., copper cables,)their properties and applications
- PC4.** examine and confirm the architecture and channel distribution of optical fiber media
- PC5.** monitor the implementation and integration of wireless communication networks
- PC6.** identify and handle various types of cables safely

### *Fiber Optical Transmission System*

To be competent, the user/individual on the job must be able to:

- PC7.** operate tools to demonstrate light transmission through fiber optics and explain its applications
- PC8.** examine optical fiber structure in metro systems for integrity
- PC9.** handle splicing tools, following the applicable safe handling procedures
- PC10.** monitor OTDR, power meters, and source meters for operational accuracy
- PC11.** execute splicing operations and validate connections using OTDR and power meters
- PC12.** set up and test SDH and PDH systems for telecommunication applications

## Qualification Pack

**PC13.** monitor SDH access and alarms, troubleshoot failures, and document corrective actions

**PC14.** configure MPLS systems for channel creation, alarm monitoring, and troubleshooting

**PC15.** Reevaluate SDH systems to MPLS for performance and scalability

### *Radio System*

To be competent, the user/individual on the job must be able to:

**PC16.** monitor the deployment and operation of wireless communication systems, including TETRA

**PC17.** perform functional tests on RAU and hand-portable systems to ensure reliability

**PC18.** operate and test train radio systems and validate sub-module functionality

**PC19.** perform functional tests of RCW and radio base station systems (BTS/MTS-4)

**PC20.** monitor the current configuration of MSO and CDRS systems for connectivity and performance

### *Telephone System*

To be competent, the user/individual on the job must be able to:

**PC21.** operate and test telephone system interfaces and connectivity in both conventional and IP-based setups

**PC22.** check and verify interconnectivity of IPBX hardware at station and OCC levels

**PC23.** execute system checks on telephone hardware used in the DMRC network

### *PIDS & PAS*

To be competent, the user/individual on the job must be able to:

**PC24.** operate and validate Passenger Information Display System (PIDS) and Public Announcement System (PAS) for real-time and scheduled information broadcasting

**PC25.** perform system checks on PIDS and PAS hardware at central and station levels

**PC26.** monitor connectivity and functionality of PIDS and PAS components

### *Closed Circuit Television System (CCTV)*

To be competent, the user/individual on the job must be able to:

**PC27.** perform functionality checks on central and station-level CCTV systems

**PC28.** perform connectivity tests to ensure proper integration of station and OCC-level CCTV systems

**PC29.** monitor CCTV hardware and troubleshoot system issues

### *Clock System*

To be competent, the user/individual on the job must be able to:

**PC30.** check synchronization of master and sub-master clock systems for operational accuracy

**PC31.** inspect and validate the interconnectivity between sub-master and slave clocks

### *Networking System*

To be competent, the user/individual on the job must be able to:

**PC32.** perform system checks on networking hardware, cables, and connectors for operational readiness

**PC33.** check synchronization of master and sub-master clock systems for operational accuracy

**PC34.** monitor and ensure interconnectivity of AFC and telecom networking hardware at station and OCC levels

**PC35.** perform practical configurations of L2 and L3 switches to meet system requirements

## Qualification Pack

### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the telecom system architecture, operations, and troubleshooting techniques
- KU2.** the properties, applications, and handling of copper and optical fiber cables
- KU3.** the tools and instruments (e.g. OTDR, power meters, splicing machines) for system installation and maintenance
- KU4.** the wireless communication protocols, radio systems, and their interconnectivity
- KU5.** the configuration and maintenance of SDH, PDH, and MPLS systems
- KU6.** the networking fundamentals, including VLANs and L2/L3 devices
- KU7.** the hardware components and connectivity of telephone, PIDS, PAS, CCTV, and clock systems
- KU8.** the safety protocols for handling telecom systems and components

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** identify and address technical issues in telecom systems
- GS2.** relay technical requirements and updates effectively to team members and stakeholders
- GS3.** evaluate system performance metrics and propose enhancements
- GS4.** ensure accuracy in configurations and operational procedures
- GS5.** record system configurations and troubleshooting steps
- GS6.** enforce safety measures during telecom operations
- GS7.** schedule and prioritize tasks effectively to meet deadlines
- GS8.** incorporate new technologies and methodologies into operations
- GS9.** guide team members in mastering operational procedures

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Telecom Systems</i>                                                                            | 2            | 2               | -             | 0.5        |
| <b>PC1.</b> interpret the telecom system architecture and communication networks                                  | -            | -               | -             | -          |
| <b>PC2.</b> familiarize with the telecom system and components through a station visit                            | -            | -               | -             | -          |
| <i>Communication Media</i>                                                                                        | 2            | 2               | -             | 1.5        |
| <b>PC3.</b> identify different types of electrical cables (e.g., copper cables,)their properties and applications | -            | -               | -             | -          |
| <b>PC4.</b> examine and confirm the architecture and channel distribution of optical fiber media                  | -            | -               | -             | -          |
| <b>PC5.</b> monitor the implementation and integration of wireless communication networks                         | -            | -               | -             | -          |
| <b>PC6.</b> identify and handle various types of cables safely                                                    | -            | -               | -             | -          |
| <i>Fiber Optical Transmission System</i>                                                                          | 5            | 5               | -             | 3          |
| <b>PC7.</b> operate tools to demonstrate light transmission through fiber optics and explain its applications     | -            | -               | -             | -          |
| <b>PC8.</b> examine optical fiber structure in metro systems for integrity                                        | -            | -               | -             | -          |
| <b>PC9.</b> handle splicing tools, following the applicable safe handling procedures                              | -            | -               | -             | -          |
| <b>PC10.</b> monitor OTDR, power meters, and source meters for operational accuracy                               | -            | -               | -             | -          |
| <b>PC11.</b> execute splicing operations and validate connections using OTDR and power meters                     | -            | -               | -             | -          |
| <b>PC12.</b> set up and test SDH and PDH systems for telecommunication applications                               | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> monitor SDH access and alarms, troubleshoot failures, and document corrective actions                                                                      | -            | -               | -             | -          |
| <b>PC14.</b> configure MPLS systems for channel creation, alarm monitoring, and troubleshooting                                                                         | -            | -               | -             | -          |
| <b>PC15.</b> Pevaluate SDH systems to MPLS for performance and scalability                                                                                              | -            | -               | -             | -          |
| <i>Radio System</i>                                                                                                                                                     | <b>2</b>     | <b>2</b>        | -             | <b>1.5</b> |
| <b>PC16.</b> monitor the deployment and operation of wireless communication systems, including TETRA                                                                    | -            | -               | -             | -          |
| <b>PC17.</b> perform functional tests on RAU and hand-portable systems to ensure reliability                                                                            | -            | -               | -             | -          |
| <b>PC18.</b> operate and test train radio systems and validate sub-module functionality                                                                                 | -            | -               | -             | -          |
| <b>PC19.</b> perform functional tests of RCW and radio base station systems (BTS/MTS-4)                                                                                 | -            | -               | -             | -          |
| <b>PC20.</b> monitor the current configuration of MSO and CDRS systems for connectivity and performance                                                                 | -            | -               | -             | -          |
| <i>Telephone System</i>                                                                                                                                                 | <b>2</b>     | <b>2</b>        | -             | <b>0.5</b> |
| <b>PC21.</b> operate and test telephone system interfaces and connectivity in both conventional and IP-based setups                                                     | -            | -               | -             | -          |
| <b>PC22.</b> check and verify interconnectivity of IPBX hardware at station and OCC levels                                                                              | -            | -               | -             | -          |
| <b>PC23.</b> execute system checks on telephone hardware used in the DMRC network                                                                                       | -            | -               | -             | -          |
| <i>PIDS &amp; PAS</i>                                                                                                                                                   | <b>2</b>     | <b>2</b>        | -             | <b>0.5</b> |
| <b>PC24.</b> operate and validate Passenger Information Display System (PIDS) and Public Announcement System (PAS) for real-time and scheduled information broadcasting | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC25.</b> perform system checks on PIDS and PAS hardware at central and station levels                          | -            | -               | -             | -          |
| <b>PC26.</b> monitor connectivity and functionality of PIDS and PAS components                                     | -            | -               | -             | -          |
| <i>Closed Circuit Television System (CCTV)</i>                                                                     | <b>2</b>     | <b>2</b>        | -             | <b>0.5</b> |
| <b>PC27.</b> perform functionality checks on central and station-level CCTV systems                                | -            | -               | -             | -          |
| <b>PC28.</b> perform connectivity tests to ensure proper integration of station and OCC-level CCTV systems         | -            | -               | -             | -          |
| <b>PC29.</b> monitor CCTV hardware and troubleshoot system issues                                                  | -            | -               | -             | -          |
| <i>Clock System</i>                                                                                                | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC30.</b> check synchronization of master and sub-master clock systems for operational accuracy                 | -            | -               | -             | -          |
| <b>PC31.</b> inspect and validate the interconnectivity between sub-master and slave clocks                        | -            | -               | -             | -          |
| <i>Networking System</i>                                                                                           | <b>2</b>     | <b>2</b>        | -             | <b>1.5</b> |
| <b>PC32.</b> perform system checks on networking hardware, cables, and connectors for operational readiness        | -            | -               | -             | -          |
| <b>PC33.</b> check synchronization of master and sub-master clock systems for operational accuracy                 | -            | -               | -             | -          |
| <b>PC34.</b> monitor and ensure interconnectivity of AFC and telecom networking hardware at station and OCC levels | -            | -               | -             | -          |
| <b>PC35.</b> perform practical configurations of L2 and L3 switches to meet system requirements                    | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                   | <b>20</b>    | <b>20</b>       | -             | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                  |
|----------------------------|------------------------------------------------------------------|
| <b>NOS Code</b>            | DMR/LSC/N0204                                                    |
| <b>NOS Name</b>            | Execute telecom systems installation, operation, and maintenance |
| <b>Sector</b>              | Logistics                                                        |
| <b>Sub-Sector</b>          |                                                                  |
| <b>Occupation</b>          | Signalling & Telecom                                             |
| <b>NSQF Level</b>          | 4.5                                                              |
| <b>Credits</b>             | 7                                                                |
| <b>Version</b>             | 1.0                                                              |
| <b>Last Reviewed Date</b>  | 08/05/2025                                                       |
| <b>Next Review Date</b>    | 08/05/2028                                                       |
| <b>NSQC Clearance Date</b> | 08/05/2025                                                       |

## Qualification Pack

# DMR/LSC/N0205: Operate and Monitor Automatic Fare Collection (AFC) Systems

## Description

This unit outlines the competencies required to operate, monitor and maintain the Automatic Fare Collection (AFC) systems, ensuring seamless performance across hardware, software, and networking components.

## Scope

The scope covers the following :

- Introduction to AFC System
- Fare Media & Business Rule
- Automatic Gates
- Ticket Office Machine (TOM)
- Ticket Issuing Machines
- Ticket Reader/Add Value Machine (TR/AVM)
- Station Computer (SC)
- Central Computer (CC) & Central Clearing House System (CCHS)
- AFC Networking
- AFC Lab Visit

## Elements and Performance Criteria

### *Introduction to AFC System*

To be competent, the user/individual on the job must be able to:

- PC1.** familiarize with AFC systems through station visits, ensuring complete understanding of system layout and functionality
- PC2.** analyze the AFC architecture, communication network, and its operational objectives

### *Fare Media & Business Rule*

To be competent, the user/individual on the job must be able to:

- PC3.** identify and differentiate types of fare media and fare products, such as tickets, NCMC, and QR code
- PC4.** demonstrate the use of NCMC and QR fare media at the Operations Control Center (OCC) and station levels
- PC5.** operate store-value and app-based ticketing systems to manage metro fare collection
- PC6.** monitor the implementation of business rules in practical scenarios and ensure compliance

### *Automatic Gates*

To be competent, the user/individual on the job must be able to:

- PC7.** monitor the operational modes and fare modes of AFC gates, ensuring alignment with passenger detection systems and design parameters
- PC8.** check gate components, including sensors and barriers, for connectivity and functionality
- PC9.** familiarize with AFC gates to understand their functionality and troubleshooting needs

## Qualification Pack

### *Ticket Office Machine (TOM)*

To be competent, the user/individual on the job must be able to:

**PC10.** check and validate the security, data storage, and integrity of TOM/EFO modules

**PC11.** familiarize with passenger information display systems and receipt printer functionality within TOM/EFO

### *Ticket Issuing Machines*

To be competent, the user/individual on the job must be able to:

**PC12.** identify the components and functionalities of ticket vending machines (TVMs), smart card vending machines, and recharge card terminal machines

**PC13.** familiarize with the operation of ticket issuing and recharge machines at stations

### *Ticket Reader/Add Value Machine (TR/AVM)*

To be competent, the user/individual on the job must be able to:

**PC14.** operate TR/AVMs for web-based recharge provisions and demonstrate their practical applications at stations

**PC15.** ensure TR/AVM functionalities comply with business rules

### *Station Computer (SC)*

To be competent, the user/individual on the job must be able to:

**PC16.** monitor data collection, status reporting, and parameter distribution through station computers

**PC17.** familiarize with SC reports, major components, and software requirements

### *Central Computer (CC) & Central Clearing House System (CCHS)*

To be competent, the user/individual on the job must be able to:

**PC18.** utilize CC/CCHS software for clock management, data distribution, and system reporting

**PC19.** familiarize with central systems at the OCC level

### *AFC Networking*

To be competent, the user/individual on the job must be able to:

**PC20.** monitor AFC network configurations and ensure proper data flow between components

**PC21.** execute practical demonstrations of AFC networking concepts

### *AFC Lab Visit*

To be competent, the user/individual on the job must be able to:

**PC22.** conduct visit to AFC lab to enhance hands-on experience with system components and troubleshooting techniques

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** AFC system architecture and operational objectives

**KU2.** AFC system communication network protocols

**KU3.** different Types of fare media (e.g., NCMC, QR, tickets)

**KU4.** the store-value ticketing systems and app-based solutions

**KU5.** Practical implementation of business rules

## Qualification Pack

- KU6.** the design features, specifications, and automatic gate components
- KU7.** the passenger detection systems and gate operational modes
- KU8.** the functionalities of TOM, TVM, and smart card vending machines
- KU9.** the data storage, security protocols, and passenger information displays
- KU10.** the data integrity, reporting mechanisms, and parameter distribution
- KU11.** the functional requirements of SC, CC, and CCHS
- KU12.** the network configurations and troubleshooting techniques in AFC networking
- KU13.** the integration of system components through networking
- KU14.** the practical handling and troubleshooting of AFC components

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** analyze system performance and troubleshoot issues effectively
- GS2.** relay technical information clearly to team members and stakeholders
- GS3.** monitor system components and identify discrepancies
- GS4.** resolve operational issues with innovative solutions
- GS5.** coordinate with the team during station visits and system demonstrations
- GS6.** adjust to new AFC technologies and fare media developments
- GS7.** ensure timely completion of tasks, such as system checks and ticketing operations
- GS8.** guide team members during practical sessions and lab visits
- GS9.** maintain safety protocols while working with AFC components

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to AFC System</i>                                                                                                                | 2            | 2               | -             | 2          |
| <b>PC1.</b> familiarize with AFC systems through station visits, ensuring complete understanding of system layout and functionality              | -            | -               | -             | -          |
| <b>PC2.</b> analyze the AFC architecture, communication network, and its operational objectives                                                  | -            | -               | -             | -          |
| <i>Fare Media &amp; Business Rule</i>                                                                                                            | 4            | 4               | -             | 3          |
| <b>PC3.</b> identify and differentiate types of fare media and fare products, such as tickets, NCMC, and QR code                                 | -            | -               | -             | -          |
| <b>PC4.</b> demonstrate the use of NCMC and QR fare media at the Operations Control Center (OCC) and station levels                              | -            | -               | -             | -          |
| <b>PC5.</b> operate store-value and app-based ticketing systems to manage metro fare collection                                                  | -            | -               | -             | -          |
| <b>PC6.</b> monitor the implementation of business rules in practical scenarios and ensure compliance                                            | -            | -               | -             | -          |
| <i>Automatic Gates</i>                                                                                                                           | 2            | 2               | -             | 1          |
| <b>PC7.</b> monitor the operational modes and fare modes of AFC gates, ensuring alignment with passenger detection systems and design parameters | -            | -               | -             | -          |
| <b>PC8.</b> check gate components, including sensors and barriers, for connectivity and functionality                                            | -            | -               | -             | -          |
| <b>PC9.</b> familiarize with AFC gates to understand their functionality and troubleshooting needs                                               | -            | -               | -             | -          |
| <i>Ticket Office Machine (TOM)</i>                                                                                                               | 2            | 2               | -             | 1          |
| <b>PC10.</b> check and validate the security, data storage, and integrity of TOM/EFO modules                                                     | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> familiarize with passenger information display systems and receipt printer functionality within TOM/EFO                                         | -            | -               | -             | -          |
| <i>Ticket Issuing Machines</i>                                                                                                                               | <b>2</b>     | <b>2</b>        | -             | <b>0.5</b> |
| <b>PC12.</b> identify the components and functionalities of ticket vending machines (TVMs), smart card vending machines, and recharge card terminal machines | -            | -               | -             | -          |
| <b>PC13.</b> familiarize with the operation of ticket issuing and recharge machines at stations                                                              | -            | -               | -             | -          |
| <i>Ticket Reader/Add Value Machine (TR/AVM)</i>                                                                                                              | <b>2</b>     | <b>2</b>        | -             | <b>0.5</b> |
| <b>PC14.</b> operate TR/AVMs for web-based recharge provisions and demonstrate their practical applications at stations                                      | -            | -               | -             | -          |
| <b>PC15.</b> ensure TR/AVM functionalities comply with business rules                                                                                        | -            | -               | -             | -          |
| <i>Station Computer (SC)</i>                                                                                                                                 | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC16.</b> monitor data collection, status reporting, and parameter distribution through station computers                                                 | -            | -               | -             | -          |
| <b>PC17.</b> familiarize with SC reports, major components, and software requirements                                                                        | -            | -               | -             | -          |
| <i>Central Computer (CC) &amp; Central Clearing House System (CCHS)</i>                                                                                      | <b>2</b>     | <b>2</b>        | -             | <b>0.5</b> |
| <b>PC18.</b> utilize CC/CCHS software for clock management, data distribution, and system reporting                                                          | -            | -               | -             | -          |
| <b>PC19.</b> familiarize with central systems at the OCC level                                                                                               | -            | -               | -             | -          |
| <i>AFC Networking</i>                                                                                                                                        | <b>2</b>     | <b>2</b>        | -             | <b>0.5</b> |
| <b>PC20.</b> monitor AFC network configurations and ensure proper data flow between components                                                               | -            | -               | -             | -          |

Logo  
Not  
Available



### Qualification Pack

| Assessment Criteria for Outcomes                                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC21.</b> execute practical demonstrations of AFC networking concepts                                                   | -            | -               | -             | -          |
| <i>AFC Lab Visit</i>                                                                                                       | <b>1</b>     | <b>1</b>        | -             | <b>0.5</b> |
| <b>PC22.</b> conduct visit to AFC lab to enhance hands-on experience with system components and troubleshooting techniques | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                           | <b>20</b>    | <b>20</b>       | -             | <b>10</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                             |
|----------------------------|-------------------------------------------------------------|
| <b>NOS Code</b>            | DMR/LSC/N0205                                               |
| <b>NOS Name</b>            | Operate and Monitor Automatic Fare Collection (AFC) Systems |
| <b>Sector</b>              | Logistics                                                   |
| <b>Sub-Sector</b>          |                                                             |
| <b>Occupation</b>          | Signalling & Telecom                                        |
| <b>NSQF Level</b>          | 4.5                                                         |
| <b>Credits</b>             | 4                                                           |
| <b>Version</b>             | 1.0                                                         |
| <b>Last Reviewed Date</b>  | 08/05/2025                                                  |
| <b>Next Review Date</b>    | 08/05/2028                                                  |
| <b>NSQC Clearance Date</b> | 08/05/2025                                                  |

## Qualification Pack

### DGT/VSQ/N0101: Employability Skills (30 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

**PC1.** understand the significance of employability skills in meeting the job requirements

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

**PC2.** identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

**PC3.** explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

**PC4.** speak with others using some basic English phrases or sentences

##### *Communication Skills*

To be competent, the user/individual on the job must be able to:

**PC5.** follow good manners while communicating with others

**PC6.** work with others in a team

## Qualification Pack

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

**PC7.** communicate and behave appropriately with all genders and PwD

**PC8.** report any issues related to sexual harassment

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

**PC9.** use various financial products and services safely and securely

**PC10.** calculate income, expenses, savings etc.

**PC11.** approach the concerned authorities for any exploitation as per legal rights and laws

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

**PC12.** operate digital devices and use its features and applications securely and safely

**PC13.** use internet and social media platforms securely and safely

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

**PC14.** identify and assess opportunities for potential business

**PC15.** identify sources for arranging money and associated financial and legal challenges

### *Customer Service*

To be competent, the user/individual on the job must be able to:

**PC16.** identify different types of customers

**PC17.** identify customer needs and address them appropriately

**PC18.** follow appropriate hygiene and grooming standards

### *Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC19.** create a basic biodata

**PC20.** search for suitable jobs and apply

**PC21.** identify and register apprenticeship opportunities as per requirement

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use basic spoken English language

**KU6.** Do and dont of effective communication

**KU7.** inclusivity and its importance

**KU8.** different types of disabilities and appropriate communication and behaviour towards PwD

**KU9.** different types of financial products and services

## Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> understand the significance of employability skills in meeting the job requirements                                                                                                                                                                | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                     | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC2.</b> identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                             | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC3.</b> explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc. | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                    | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC4.</b> speak with others using some basic English phrases or sentences                                                                                                                                                                                    | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC5.</b> follow good manners while communicating with others                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC6.</b> work with others in a team                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                                                                                                                                                               | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC7.</b> communicate and behave appropriately with all genders and PwD                                                                                                                                                                                      | -            | -               | -             | -          |
| <b>PC8.</b> report any issues related to sexual harassment                                                                                                                                                                                                     | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                                                                                                                                                            | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC9.</b> use various financial products and services safely and securely                                                                                                                                                                                    | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> calculate income, expenses, savings etc.                                             | -            | -               | -             | -          |
| <b>PC11.</b> approach the concerned authorities for any exploitation as per legal rights and laws | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                   | <b>4</b>     | <b>6</b>        | -             | -          |
| <b>PC12.</b> operate digital devices and use its features and applications securely and safely    | -            | -               | -             | -          |
| <b>PC13.</b> use internet and social media platforms securely and safely                          | -            | -               | -             | -          |
| <i>Entrepreneurship</i>                                                                           | <b>3</b>     | <b>5</b>        | -             | -          |
| <b>PC14.</b> identify and assess opportunities for potential business                             | -            | -               | -             | -          |
| <b>PC15.</b> identify sources for arranging money and associated financial and legal challenges   | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC16.</b> identify different types of customers                                                | -            | -               | -             | -          |
| <b>PC17.</b> identify customer needs and address them appropriately                               | -            | -               | -             | -          |
| <b>PC18.</b> follow appropriate hygiene and grooming standards                                    | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC19.</b> create a basic biodata                                                               | -            | -               | -             | -          |
| <b>PC20.</b> search for suitable jobs and apply                                                   | -            | -               | -             | -          |
| <b>PC21.</b> identify and register apprenticeship opportunities as per requirement                | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                  | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0101                   |
| <b>NOS Name</b>            | Employability Skills (30 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 2                               |
| <b>Credits</b>             | 1                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 12/03/2026                      |
| <b>Next Review Date</b>    | 12/03/2029                      |
| <b>NSQC Clearance Date</b> | 12/03/2026                      |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on a knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per the assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practicals for every student at each examination/training centre based on this criterion.
6. To pass the Qualification Pack, every trainee should score a minimum of 60% of aggregate marks to successfully clear the assessment.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

## Qualification Pack

### Minimum Aggregate Passing % at QP Level : 60

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

### Assessment Weightage

Compulsory NOS

| National Occupational Standards                                                | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|--------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| DMR/LSC/N9901.Implement metro operations training and safety procedures        | 20           | 20              | 0             | 10         | 50          | 10         |
| DMR/LSC/N0203.Conduct basic signalling operations                              | 20           | 20              | 0             | 10         | 50          | 30         |
| DMR/LSC/N0204.Execute telecom systems installation, operation, and maintenance | 20           | 20              | 0             | 10         | 50          | 25         |
| DMR/LSC/N0205.Operate and Monitor Automatic Fare Collection (AFC) Systems      | 20           | 20              | 0             | 10         | 50          | 25         |
| DGT/VSQ/N0101.Employability Skills (30 Hours)                                  | 20           | 30              | -             | -          | 50          | 10         |
| <b>Total</b>                                                                   | <b>100</b>   | <b>110</b>      | <b>-</b>      | <b>40</b>  | <b>250</b>  | <b>100</b> |



## Qualification Pack

### Acronyms

|                |                                                        |
|----------------|--------------------------------------------------------|
| <b>NOS</b>     | National Occupational Standard(s)                      |
| <b>NSQF</b>    | National Skills Qualifications Framework               |
| <b>QP</b>      | Qualifications Pack                                    |
| <b>TVET</b>    | Technical and Vocational Education and Training        |
| <b>NCVET</b>   | National Council for Vocational Education and Training |
| <b>QP</b>      | Qualification Pack                                     |
| <b>MC</b>      | Model Curriculum                                       |
| <b>NSQF</b>    | National Skills Qualification Framework                |
| <b>NSQC</b>    | National Skills Qualification Committee                |
| <b>NOS</b>     | National Occupational Standards                        |
| <b>NCO</b>     | National Classification of Occupations                 |
| <b>ES</b>      | Employability Skills                                   |
| <b>MR</b>      | Metro Railway                                          |
| <b>DMRC</b>    | Delhi Metro Rail Corporation                           |
| <b>DMRA</b>    | Delhi Metro Rail Academy                               |
| <b>AFC</b>     | Automatic Fare Collection                              |
| <b>ATS</b>     | Automatic Train Supervision                            |
| <b>ATP</b>     | Automatic Train Protection                             |
| <b>ATP</b>     | Automatic Train Protection                             |
| <b>S&amp;T</b> | Signalling and Telecommunications                      |
| <b>CBTC</b>    | Communications-Based Train Control                     |
| <b>RRI</b>     | Relay-Interlocking                                     |
| <b>SSI</b>     | Solid State Interlocking                               |
| <b>CBI</b>     | Computer-Based Interlocking                            |
| <b>PSD</b>     | Platform Screen Doors                                  |

## Qualification Pack

|                |                                           |
|----------------|-------------------------------------------|
| <b>E&amp;M</b> | Electrical and Mechanical                 |
| <b>OHE</b>     | Overhead Electrification                  |
| <b>SCADA</b>   | Supervisory Control and Data Acquisition  |
| <b>TVS</b>     | Train Vision System                       |
| <b>OTDR</b>    | Optical Time Domain Reflectometer         |
| <b>SDH</b>     | Synchronous Digital Hierarchy             |
| <b>PDH</b>     | Plesiochronous Digital Hierarchy          |
| <b>MPLS</b>    | Multiprotocol Label Switching             |
| <b>PIDS</b>    | Passenger Information Display System      |
| <b>PAS</b>     | Public Announcement System                |
| <b>IPBX</b>    | Internet Protocol Private Branch Exchange |
| <b>OCC</b>     | Operations Control Center                 |
| <b>NCMC</b>    | National Common Mobility Card             |
| <b>TVMs</b>    | Ticket Vending Machines                   |
| <b>TOM</b>     | Ticket Office Machine                     |
| <b>EFO</b>     | Electronic Fare Organizer                 |
| <b>TR/AVR</b>  | Ticket Reader/Add Value Machine.          |
| <b>SC/CC</b>   | Station Computer/ Central Computer        |
| <b>CCHS</b>    | Central Clearing House System             |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |